

Chris Jiang 400322193
Sohel Saini 400327615

Exercise 1:

In exercise 1 we used approximately 22 registers in our design. 2 bits were for the FSM state to manage the control flow, we used 9 bits each for the read and write addresses to access the memory locations, and 1 bit each for the write enable signals `write_enable_b[0]` and `write_enable_b[1]`. This estimate was almost the same as the Quartus compilation report which reported 23 registers. The `read_data_a[0]` and `read_data_a[1]` are assigned to signals and they were concatenated with the MSB and the signal to increase the signal by 1 bit. An if first checks if it's in the top half or bottom half of memory and then the signed bit is checked to see what equations to perform. The results are then written back into the memory at `write_data_b[0]` and `write_data_b[1]`.